
S2G-RAG PILOT EVALUATION REPORT

Retrieval effectiveness, evidence coverage, answer quality, latency, cost, and reproducible evaluation protocol

PROVISIONAL - AGENT-ASSISTED PILOT
Not a frozen thesis benchmark until independent human verification is complete.

Corpus	Canonical Corpus V1.1 - 27 PDFs - 3,482 chunks
Pilot	20 queries - 27 evidence groups - 23 unique GT chunks
Runtime	20/20 completed - 89 GPT-5 nano responses - 0 infrastructure failures
Document date	2026-09-13

Part I - Pilot Evaluation Results

Observed results from the completed 20-query engineering pilot. All caveats and validity limits are retained.

Status

Execution: PASS (20/20 queries completed). This is an engineering pilot, not yet a frozen thesis benchmark. The benchmark contains agent-assisted source verification and **0/20 independently human-verified queries**. The strict official validator therefore fails by design. Results below must be labelled *provisional* until `ground_truth_review.csv` is completed.

- Corpus: Canonical Corpus V1.1, 27 PDFs, 3,482 indexed chunks.
- Corpus SHA-256: `fd2358332668de3dca8c099ff6a3dd0cf1d23b47c378f7c92835df4a08c34c02`.
- Pilot: 20 queries, 27 required evidence groups, 23 unique GT chunks.
- Distribution: 5 direct lookup, 5 paraphrase, 3 terminology mismatch, 4 multi-evidence, 3 cross-document.
- Runtime GT firewall: PASS. Runtime input contains only `id` and `query`.
- Model: requested `gpt-5-nano`; all 89 responses resolved to `gpt-5-nano-2025-08-07`; all response IDs were present and unique.
- API/infrastructure failures: 0.

Metric definitions

- Recall@k is the proportion of unique GT chunks found in top-k, macro-averaged.
- MRR@10 uses the rank of the first GT chunk.
- Evidence-group coverage (EGC)@k is the proportion of required evidence groups with at least one acceptable chunk in top-k.
- Complete evidence is achieved only when every group for a query is covered.

Every pilot evidence group currently has one acceptable chunk, so Recall@k and EGC@k are numerically equal. They remain separate implementations because future groups may contain multiple acceptable chunks.

B0 retrieval results

Stage	R@1	R@3	R@5	R@10	MRR@10	EGC@10	Complete @10
BM25	0.525	0.800	0.850	0.875	0.746	0.875	16/20
Dense	0.500	0.625	0.775	0.850	0.672	0.850	16/20
Hybrid + RRF	0.525	0.825	0.825	0.875	0.742	0.875	16/20
Hybrid + RRF + reranker	0.575	0.825	0.825	0.825	0.767	0.825	15/20

The reranker improves rank-one performance and MRR over raw hybrid, but it pushes one query's complete evidence out of top-10 (16 to 15). This is a measured trade-off, not evidence for tuning on the same pilot.

For the final B0 stage, direct lookup is 1.000 R@10, terminology mismatch is 1.000, paraphrase is 0.800, cross-document is 0.667, and multi-evidence is 0.625. Only 2/4 multi-evidence queries and 1/3 cross-document queries have complete evidence at top-10.

S2G retrieval/evidence results

The candidate and final-evidence layers are reported separately. The iterative candidate pool contains the top six reranked chunks from each executed retrieval round. Selected evidence contains parent chunks admitted by the sentence selector; answer citations are the subset used by the answer reasoner.

Stage	R@1	R@3	R@5	R@10	MRR@10	EGC@10	Complete @10
Iterative candidate pool	0.625	0.750	0.800	0.800	0.777	0.800	14/20
S2G selected evidence	0.625	0.700	0.750	0.750	0.785	0.750	13/20
Answer citations	0.600	0.750	0.750	0.750	0.767	0.750	13/20

S2G does not outperform B0 top-10 retrieval on this pilot. Compared with B0 hybrid+reranker, S2G candidate R@10 changes from 0.825 to 0.800, and selected/cited R@10 falls to 0.750. Its selected-evidence MRR@10 is slightly higher (0.785 versus 0.767), showing stronger early precision for recovered evidence but weaker total coverage.

The main failure is multi-evidence coverage. At the selected-evidence stage its R/EGC@10 is 0.250 and complete evidence is 0/4. Direct lookup and terminology mismatch are both 1.000; paraphrase is 0.800; cross-document is 0.667.

Evidence-flow diagnosis

- P007 and P015: GT is absent from both B0 top-10 and the S2G candidate pool.
- P014 and P018-P019: only one of two required groups is recovered/cited.
- P016: both GT chunks are present in the iterative candidate pool, but neither is admitted to selected evidence. This is specifically a selector/controller loss, not a first-stage retrieval miss.
- P017: B0 top-10 contains both pilot GT chunks, while the S2G top-six-per-round candidate pool contains one. Human review also found that the question has a possible ground-truth scope ambiguity between a general and a specific rule.

No retrieval parameter or prompt was changed after observing these cases.

Agent-assisted answer-quality review

This review is useful for debugging but is **not an independent human evaluation**. PASS rates below use all 20 queries as the denominator; REVIEW is not counted as PASS.

Criterion	PASS	FAIL	REVIEW	PASS / 20
Correctness	14	5	1	70%
Completeness	13	6	1	65%
Faithfulness	15	4	1	75%
Citation adequacy	12	7	1	60%
Clarity	18	2	0	90%

Ten of 20 answers pass all four core criteria (correctness, completeness, faithfulness, citation adequacy). Dominant problems are failure to answer all parts of multi-evidence questions, citing only one required document/group, and answering from a plausible but wrong nearby provision. P013 incorrectly implies that possessing a MOS certificate alone is enough to skip the course; the source requires an assessment and threshold. P015 is materially wrong, P016 declares insufficient, and P019 omits the undergraduate half. P017 remains REVIEW because two applicable source provisions express timing at different specificity levels.

The untouched human worksheet is `outputs/evaluation/s2g_pilot_v1/answer_quality_review.csv`.

Latency, tokens, and API cost

B0 local retrieval latency (seconds/query)

Component	Mean	Median	P95	Max
BM25	0.011	0.010	0.020	0.022
Dense	0.088	0.062	0.131	0.766
Reranker	0.999	0.956	1.215	1.913
Total	1.099	1.017	1.368	2.694

The B0 API cost is zero because its retrieval models ran locally. This does not include hardware, electricity, or engineering cost.

S2G resources

- End-to-end latency: mean 22.978 s, median 21.666 s, P95 41.059 s, maximum 45.716 s.
- Provider-call latency: mean 20.895 s/query.
- Retrieval latency across iterative rounds: mean 2.046 s/query.
- Retrieval rounds: mean 1.30, median 1, maximum 3.
- Stop reasons: 17 STOP_SUFFICIENT, 3 STOP_NO_NEW_EVIDENCE.
- Provider calls: 89 total, 4.45/query: 43 judge, 26 sentence selector, and 20 answer reasoner calls.
- Tokens: 143,847 input; 65,214 output, including 46,720 reasoning tokens; zero cached input tokens were reported.
- Estimated OpenAI API cost: **USD 0.03327795 total**, or **USD 0.00166390/query**.

The estimate uses the recorded token usage and the GPT-5 nano prices observed on 2026-09-13: USD 0.05/M input tokens, USD 0.005/M cached input tokens, and USD 0.40/M output tokens. It excludes local compute and network costs.

Validity decision and next checkpoint

The software and runtime execution pass; the evidence shows the deployed S2G system works well on direct/terminology questions but is not yet reliable for multi-evidence answers. These results are suitable for engineering diagnosis and a progress update. They are **not yet suitable as final thesis claims**, for two reasons:

1. the benchmark has not received independent human ground-truth verification;
2. answer quality was agent-assisted, and P017 exposes at least one annotation ambiguity.

The next step is to review every row in `ground_truth_review.csv`, resolve P017 without looking at retrieval rankings, and freeze the benchmark. Then run a fresh, untuned official experiment and have answer quality reviewed independently. Do not change retrieval/controller parameters on the basis of this pilot and reuse the same questions as a supposedly unbiased test.

Appendix A - Evaluation Protocol

Definitions, ground-truth firewall, reproducible commands, human-review requirements, and interpretation limits.

Scope and status

This protocol measures the current full-PDF product on five dimensions:

1. retrieval recall;
2. reciprocal rank (MRR);
3. evidence-group coverage (EGC);
4. human-reviewed answer quality;
5. end-to-end latency and OpenAI API cost.

`data/evaluation/s2g_pilot_v1/queries.json` is a 20-query, 27-evidence-group pilot grounded in Canonical Corpus V1.1. It is currently `DRAFT_AGENT_VERIFIED_PENDING_HUMAN`. The raw scores may be used for engineering diagnosis, but must not be described as results from an independent human-annotated study. Official thesis reporting requires completion of `ground_truth_review.csv` and changing each verification status only after review.

Ground-truth firewall

The benchmark file contains evidence labels. The runtime export contains exactly `id` and `query`; loaders reject every additional field. Retrieval and S2G execution therefore cannot inspect ground truth. The offline evaluator loads evidence labels only after rankings/results have been finalized.

The default CLI requires `human_verified`. `--allow-pending-human` is an explicit opt-in for provisional engineering measurements.

Retrieval metrics

- `Recall@k`: unique relevant chunk IDs retrieved in the first `k`, divided by all unique relevant chunk IDs for the query.
- `MRR@10`: reciprocal rank of the first relevant chunk in the first 10; zero when none is present.
- `Evidence-group coverage@k`: required evidence groups covered in the first `k`, divided by the number of required groups. A group is covered when at least one of its acceptable chunk IDs appears.
- `Complete-evidence@k`: true only when every required group is covered.

Metrics are reported per query, as macro averages, and by query type. For S2G the report deliberately separates the iterative candidate pool, chunks presented by the reranker, sentence-selected parent chunks, and answer citations. These are not interchangeable stages.

Answer quality

Automated checks cover only enforceable contracts: citation IDs are drawn from final evidence and a substantive answer has citations. Semantic quality is reviewed in `answer_quality_review.csv`. A reviewer records TRUE/FALSE for correctness, completeness, faithfulness to evidence, citation adequacy, and clarity, plus notes. All HUMAN fields are blank when generated. Report both numerator/denominator and percentage; do not silently exclude insufficient answers.

Latency and cost

Retrieval traces contain BM25, dense, fusion, reranker, and total latency. S2G traces contain per-retrieval latency, provider-call latency, and end-to-end pipeline latency. Reports use mean, median, P95, and maximum across queries.

Token cost uses recorded Responses API usage (input_tokens, cached input tokens, and output_tokens). The configuration records the observed price date and source. The estimate excludes local compute, electricity, and network costs and must be reported as an API estimate, not total system cost.

Reproducible commands

Use the project environment:

```
[bash]
conda run -n ml-env python -m src.evaluation.cli validate
conda run -n ml-env python -m src.evaluation.cli export-runtime \
  --output data/evaluation/s2g_pilot_v1/runtime_queries.json
conda run -n ml-env python -m src.evaluation.cli run-b0 \
  --runtime-queries data/evaluation/s2g_pilot_v1/runtime_queries.json \
  --output outputs/evaluation/s2g_pilot_v1/b0_rankings.json
conda run -n ml-env python -m src.evaluation.cli evaluate-b0 \
  --rankings outputs/evaluation/s2g_pilot_v1/b0_rankings.json \
  --output outputs/evaluation/s2g_pilot_v1/b0_metrics.json
conda run -n ml-env python -m src.evaluation.cli run-s2g \
  --runtime-queries data/evaluation/s2g_pilot_v1/runtime_queries.json \
  --output-root outputs/evaluation/s2g_pilot_v1/s2g_runtime
conda run -n ml-env python -m src.evaluation.cli evaluate-s2g \
  --runtime-root outputs/evaluation/s2g_pilot_v1/s2g_runtime \
  --output-root outputs/evaluation/s2g_pilot_v1
```

Until human verification is complete, append `--allow-pending-human` only to validation, export, and offline evaluation commands. Execution commands consume the already-sanitized runtime file and never load the benchmark.

Interpretation limits

Twenty stratified questions are a pilot, not a comprehensive estimate of all possible student questions. Query authoring, ground-truth review, model behavior, OCR noise, and document lifecycle ambiguity are separate threats to validity. No prompt, retrieval setting, or query may be tuned after inspecting individual pilot outcomes and then reported on the same set as an unbiased result.

Source documents: docs/s2g-pilot-evaluation-results.md and docs/evaluation-protocol.md. Generated reproducibly without modifying either source file.